Part 22 — Strain stability as one chain, ALE as a designed campaign, the screening funnel, and the tank that crashes

A reviewer made two criticisms of this module: that strain stability is present but not integrated with scale-up, and that the analytical bottleneck behind high-throughput screening is never costed. Both check out against an audit. I added a third that neither of us had covered and that ends more campaigns than either.

What the audit found

Topic	Mentions	Verdict
Plasmid loss and retention arithmetic	26	Present (PP3, part 9 ch 32)
Cheater and revertant takeover	32	Present (Z8)
Mutation supply at scale	14	Present (part 14 ch B10)
IS elements and transposon activity	6	**Thin**
Hypermutators and mismatch repair	4	**Thin**
ALE as a technique	100	Well referenced
ALE as a designed campaign (population genetics)	—	**Absent as arithmetic**
Assay throughput in samples per day	**0**	**Absent**
The screening funnel / tiered assays	**0**	**Absent**
Contamination and phage, quantified	—	**Absent — my addition**

So the reviewer was right twice. The components of stability were all there and never multiplied together; and FACS throughput and chromatography throughput both appear in the module without their ratio ever being computed, which is precisely where the industrial friction sits. Four chapters, Y1 to Y4, and eight problems, SB1 to SB4 and SC1 to SC4, taking the set to **149**.

Chapter Y1 · integrates PP3, Z8 and part 14 ch B10

The stability chain: one calculation from vial to harvest

Each mechanism was already in the module. What follows is what happens when you multiply them along a real seed train — and the answer is that batch processes are safe and repeated-batch ones are not.

Stage by stage, with the generations counted

A seed train from a cryovial to a 100 m³ vessel is not one 27-generation step; it is five expansions plus a production phase, and plasmid loss compounds through all of them. With a loss rate of 5×10^{-4} per division and an 8% growth advantage for the cell that lost it:

Stage	Cumulative generations	Plasmid-positive fraction
Vial to 1 mL	0	1.000
1 mL to 50 mL	5.6	0.996
50 mL to 5 L	12.2	0.988
5 L to 500 L	18.8	0.976
500 L to 100 m³	26.4	**0.951**
Through the production phase	34.4	**0.907**

Two things follow that a single-step calculation hides. The strain is **91% plasmid-positive at harvest, not the 95% you would measure at the end of the seed train** — and the seed-train figure is the one most groups measure, because it is the last point before the big tank. And because biomass doubles every generation, **the last three generations contain 88% of all the cells**, so the loss is concentrated exactly where the product is made. The cell-weighted average plasmid content across the whole train is **0.875**, which is the number that actually multiplies your yield.

The mechanism the plasmid assay cannot see

Plating with and without antibiotic measures *plasmid presence*. It does not measure whether the pathway on the plasmid still works. The dominant route to a silently broken pathway in *E. coli* is **insertion-sequence transposition**: an IS element hops into a burdensome gene, the cell keeps the plasmid and its resistance marker, and it grows faster than its producing siblings. Such a cell is antibiotic-resistant, plasmid-positive, and useless.

The arithmetic decides when to worry. Without selection, an IS insertion arising at 10^{-6} per cell per generation reaches only **4×10^{-5} after 40 generations** and 4×10^{-4} after 400 — negligible. With an 8% growth advantage it still reaches only 10^{-11} by generation 40 from a single founder. But the time to sweep from 10^{-6} to half the population is

Growth advantage	Generations to 50%	Days at 10 generations/day
5%	283	28
8%	180	18
15%	99	10

So the conclusion is sharp and operationally useful: **a 40-generation batch train is safe, and anything running for hundreds of generations is not**. Repeated-batch operation makes the point concretely — re-inoculating from the previous batch, the plasmid-positive fraction falls from 0.876 after one batch to **0.233 after two and 0.001 after four**. The practice of carrying a culture forward to save a seed train destroys the strain in three passages, and it does so invisibly if you only assay the first batch.

Hypermutators: an asset in evolution, a liability in production

A *mutS* or *mutL* defect raises the point-mutation rate 30- to 100-fold. In a 100 m³ fermenter at 10^{16} cells, the number of new inactivating mutations arising in a single 1 kb pathway gene per generation goes from **10^{10} to 3×10^{11} at 30× and 10^{12} at 100×**. Those numbers are already enormous at the wild-type rate, which is part 14 chapter B10's point; what the hypermutator changes is the *supply of double mutants and of rarer events*, including the IS and deletion events that break pathways.

Hence the rule: a hypermutator is **deliberately useful in an ALE campaign**, where you want mutation supply, and must be **repaired before the strain goes to production**, where you want genetic stasis. Many published evolved strains carry mismatch-repair defects acquired during evolution, and checking for them in the sequencing output is part of characterising an evolved isolate.

Seed train	five expansions + production = 34 generations; 0.951 → 0.907 plasmid-positive
Weighting	last 3 generations hold 88% of cells; cell-weighted plasmid content 0.875
IS sweep	180 generations from 10^{-6} to 50% at an 8% advantage; batch safe, repeated batch not
Repeated batch	0.876 → 0.233 → 0.001 over four re-inoculations
Hypermutator	30–100× rate; asset in ALE, liability in production

Chapter Y2 · extends part 11 and problem Y6

Adaptive laboratory evolution as a designed campaign

The module referenced ALE a hundred times and never computed a campaign. These are the four numbers that design one: mutation supply, bottleneck size, fixation time, and how many lineages you need to tell cause from passenger.

Mutation supply: why ALE is a selection experiment

The number of new genome mutations entering a population each generation is $N_e \times \mu \times$ genome size. For *E. coli* at 2×10^{-9} per base per generation and 4.64 Mb:

Effective population	New genome mutations per generation	Times every base is hit
10^7	9.3×10^4	0.02
10^9	9.3×10^6	**2.0**
10^{11}	9.3×10^8	200

At $N_e \geq 10^9$ — a litre of culture at modest density — **every possible point mutation in the genome arises every generation**. That reframes the whole method: ALE is not a mutation experiment, it is a **selection experiment**, and the design variables are the selection pressure and the population structure, not the mutagenesis. Chemical mutagenesis or a hypermutator adds value only for multi-mutation events or at small population size.

Bottlenecks, time, and what a transfer regime costs

Serial transfer at 1:100 gives $\log_2(100) =$ **6.6 generations per cycle**, so a 500-generation campaign is **75 daily cycles — about eleven weeks**; at 1:1000 it is 10 generations per cycle and 50 days. The deeper dilution is faster in calendar time and *harsher* in population genetics, because each transfer is a bottleneck that discards rare beneficial lineages.

Fixation time for a beneficial mutation of advantage s, from a single cell in $N_e = 10^9$:

Advantage s	Generations to fixation	Probability of escaping drift (≈2s)
2%	1,036	4%
5%	414	10%
10%	207	20%
30%	69	60%

Two design consequences. A 2% improvement needs **more than a thousand generations** to fix and has a 96% chance of being lost to drift before it starts — so an ALE campaign of a few hundred generations can only find mutations worth about 5% or more, and if your objective offers only small gains, ALE is the wrong tool. And because mutation supply at 10^9 is enormous, **several beneficial lineages compete simultaneously** — clonal interference — so the mutation that fixes is the best of many contenders rather than the first to arise, which is why parallel replicate lineages often fix *different* mutations in the same gene.

Telling the causal mutation from the passengers

An evolved isolate typically carries several to tens of mutations, most of them hitchhikers that rose with a beneficial one. The statistical answer is **parallel independent lineages**. If a given passenger appears in any one lineage with probability 0.3, the chance it appears in all of n lineages is 0.3^n:

Independent lineages	Chance a passenger appears in all
3	2.7×10^{-2}
6	7.3×10^{-4}
12	**5.3×10^{-7}**

So a gene hit in all of six to twelve independent lineages is causal with high confidence, and that — not the number of generations — is what the experimental design should buy. **Run 6 to 12 parallel lineages rather than one deep one.** The confirmation remains reconstruction: put the candidate mutation into the clean parent and measure, as part 8 chapter 28 required, because parallelism establishes association and reconstruction establishes cause.

Mutation supply	$N_e\mu L$: every base hit twice per generation at $N_e = 10^9$
Transfer regime	1:100 = 6.6 generations/cycle; 500 generations ≈ 75 days
Fixation	$\ln(N_e)/s$ generations; 2s chance of escaping drift
Resolution limit	a few hundred generations finds s ≥ 5% only
Causality	6–12 parallel lineages, then reconstruct in the parent

Chapter Y3 · extends weeks 29, 30 and 34

The screening funnel: the analytical bottleneck is the design constraint

The module taught sorting at 10^7 and chromatography at 110 samples a day in different chapters and never divided one by the other. The ratio is the single most useful number in campaign design.

What each assay actually delivers in a working day

Assay	Per working day	Unit	What it measures
Sorting or selection	5×10^7	cells	A fluorescence proxy
Microtitre optical screen	3,000	clones	A colour, fluorescence or growth proxy
LC-MS with sample prep	**150**	samples	**The actual product**
HPLC, 12-minute method	110	samples	The actual product
Shake flask plus HPLC	24	strains	Product under growth conditions
Bench bioreactor	4	strains	Product under process conditions

The mismatch is seven orders of magnitude from end to end, and the one that bites is in the middle: a sorter hands you **10^5 candidates** and mass spectrometry verifies **150 a day**, a ratio of about **670 to 1**. You cannot verify what you sorted. That is not a resourcing problem to be fixed with another instrument; it is a structural feature, and the response is to design the campaign as a funnel whose tiers are sized by the capacity of the assay below them.

A funnel designed backwards from capacity

Tier	Input	Kept	Output	Instrument-days
Sort or select	10^7	0.1%	10,000	0.2
Optical plate screen	10^4	10%	1,000	3.3
HPLC confirmation	300	30%	90	2.7
Flask and bioreactor	20	—	20	0.8

Seven instrument-days in total, and note the structure: the tier that *sounds* most impressive, the sort, costs 0.2 days, while the optical screen costs 3.3 and the chromatography 2.7. **The cheap tier is the big one, and the expensive tiers are small by necessity.** Note also the discard between tiers two and three: 1,000 clones pass the optical screen and only 300 go to HPLC, which means the ranking from the proxy assay is being trusted to throw away 700 — and that is where a campaign silently loses its best strain.

Where the mass-spec time actually goes

Step, per 96-well plate	Hours
Quench and extract	2.0
Dry down	1.5
Derivatise (for GC-MS)	2.0
Resuspend and plate	0.5
Instrument run at 12 min per sample	**19.2**
Total	**25.2 h**

One plate is **25 hours end to end**, so a single operator with one instrument manages about **1.6 plates a week** — roughly 150 samples. That is the number the reviewer was pointing at, and it explains three industrial habits that look like laziness from outside and are arithmetic from inside: **shortening the chromatographic method** to two or three minutes, accepting worse separation for six times the throughput; **pooling** four wells into one injection, which multiplies throughput fourfold and destroys the identity of three in four wells unless you deconvolve; and **building an optical proxy for everything**, with all the sensor-gaming risk that EV5 quantified.

The design rule that falls out: **decide the bottom tier first**. Count how many true measurements you can make, multiply upward by each tier's kept fraction, and that tells you how large a library is worth building. Building 10^7 variants when the bottom tier is 20 bioreactor runs is not ambition; it is a funnel whose top is detached from its bottom.

The ratio	sort 10^5 candidates/day against MS 150/day = 670:1
MS reality	25.2 h per 96-well plate; ~1.6 plates per operator-week
Funnel	10^7 → 10^4 → 300 → 20 over about seven instrument-days
Design rule	size the library from the bottom tier upward, not the top down

Chapter Y4 · extends part 14 ch B10 · my addition

Contamination and phage: the failure that ends campaigns

Part 1 chapter 10 computed sterilisation and then said that filtration does not stop phage, without computing what that means. This is the chapter that asks what happens when tank three crashes on day four.

Contamination as a per-batch probability

Sterilisation arithmetic (part 14 chapter B10) answers "did we kill what was in the medium". Operational contamination is a different question — a seal, a sample port, a drain, an air filter — and it is best treated as a per-batch probability:

Per-batch probability	Expected losses in 50 batches	P(no loss at all)
2%	1.0	0.364
5%	2.5	0.077
10%	5.0	**0.005**

Read the third column. At a 5% contamination rate — not unusual for a poorly characterised process — the probability of completing a 50-batch campaign without a loss is **8%**, so you should plan for losses rather than hope. The planning consequence is V1's and X24's: build the expected losses into the schedule as extra batches, not as an optimistic assumption, and the economic consequence is that each loss carries the full cost of the medium, the vessel time and the seed train.

Phage, computed

Phage is worse than bacterial contamination for a reason that is arithmetic rather than mysterious: it replicates on your product organism, with a burst size near 100 and a latent period near 30 minutes. A single infective particle multiplies by 100 every half hour, so the time to sweep a population is $\ln(1/f_0)/(\ln(\text{burst})/\text{latent})$:

Starting free-phage fraction	Time to collapse
10^{-9}	**2.2 h**
10^{-7}	1.8 h

Against a 48-hour fermentation, **a phage entering at inoculation ends the run before the first shift change**, and the operational signature is unmistakable once you know it: optical density falls, the dissolved oxygen trace rises sharply as respiration stops, and the broth clears. That rising-DO signature is the same one problem X6 attributed to substrate inhibition, which is why the *first* diagnostic is a microscope or a plaque assay rather than a feed adjustment.

What actually works, in order

Defence	What it does	Limit
Do not filter and hope	A 0.2 µm filter removes bacteria; phage are 30–200 nm and pass	Needs 0.02 µm filtration or heat for any phage-risk stream
Segregate	Separate air, drains and personnel flow between seed train and production	Most plant outbreaks trace to a drain or a leaking seal, not the medium
Rotate strains	Different receptor profiles, so one phage cannot take the whole plant	Needs two qualified strains, which is a real development cost
CRISPR immunity	Spacers against the local phage population in the production strain	Phage escape by protospacer mutation; needs re-spacering
Receptor deletion	Delete the surface protein the phage uses	Often costs fitness, and selects for phage using another receptor
Lysogen screening	Check the production strain for inducible prophage before scale-up	A resident prophage can be induced by stress and is self-inflicted

The one most worth knowing for an interview: **sequence your production strain for prophages before you scale it**. A strain that carries an inducible prophage will crash under the stress of a large fermenter and the cause will look like contamination from outside.

Contamination	$(1-p)^n$: at 5% per batch, only 8% chance of 50 clean batches
Phage sweep	$\ln(1/f_0)/(\ln \text{burst/latent}) \approx 2$ h from 10^{-9}
Signature	falling OD, rising dissolved oxygen, clearing broth
Filtration	0.2 µm stops bacteria, not 30–200 nm phage

Problems SB1–SB4 and SC1–SC4

SB1 · chapter Y1 · 2 h

Carry plasmid loss along the whole seed train

1. Count the generations for each expansion from a vial to 100 m³, and compute retention stage by stage at a loss rate of 5 × 10^{-4} and an 8% burden.
2. What fraction is plasmid-positive at harvest, and how does that differ from the end-of-seed-train figure most groups measure?
3. What fraction of all cells exists in the last three generations, and what is the cell-weighted plasmid content?
4. Repeat for four rounds of repeated-batch re-inoculation.

Worked solution. Retention falls 1.000 → 0.996 → 0.988 → 0.976 → **0.951** at the end of the seed train, and **0.907** after the production phase. So the harvest figure is four points below the one usually measured, and in the wrong direction. Because biomass doubles each generation, **the last three generations hold 88% of the cells**, so the loss is concentrated where production happens; the cell-weighted plasmid content across the train is **0.875**, which is the figure that multiplies your titer.

Repeated batch is the striking case: re-inoculating from the previous culture gives **0.876, 0.233, then 0.001 by the fourth round**. Carrying a culture forward to save a seed train destroys the strain in three passages — and if you assay only the first batch, the collapse is invisible until the yield goes.

What it teaches: plasmid loss must be integrated along the whole train and weighted by where the cells are, and the standard end-of-seed-train assay measures the wrong point.

SB2 · chapter Y1 · 1.5 h

The reversion the antibiotic plate cannot see

1. Compute the fraction carrying an IS insertion in the pathway after 40, 200 and 400 generations with no selection, at rates of 10^{-7}, 10^{-6} and 10^{-5}.
2. Compute the generations for such a mutant to go from 10^{-6} to 50% at advantages of 5%, 8% and 15%.
3. Which operating modes are safe and which are not?
4. Why does plating with and without antibiotic miss this entirely, and what assay would catch it?

Worked solution. Without selection the frequencies stay small — **4 × 10^{-5} at 10^{-6} and 40 generations**, and only 4 × 10^{-3} at the highest rate and 400 generations. With selection the sweep times are **283, 180 and 99 generations** at the three advantages, which is **28, 18 and 10 days** at ten generations a day.

So: a 40-generation batch train is **safe**; repeated batch over hundreds of generations is **not**; continuous culture certainly is not, which is PP9's and Z8's conclusion arrived at by a third route. The antibiotic plate misses it because the cell *still has the plasmid and the marker* — only the pathway is broken. What catches it: **product per cell**, measured on individual colonies rather than the population; a **functional reporter** in the pathway; or **sequencing the construct** from end-of-run isolates, where an IS insertion appears as a new junction in exactly the evidence table problem Y6 told you to read.

What it teaches: genetic stability has two independent failure modes, and the routine assay detects only one of them.

SB3 · chapter Y2 · 2 h

Design an ALE campaign

1. Compute the new genome mutations per generation at effective populations of 10^7, 10^9 and 10^{11}, and say how many times each base is hit.
2. What does that imply about whether to add a mutagen or a hypermutator?
3. Compute generations per cycle at 1:100 and 1:1000 transfer, and the calendar time for 500 generations.
4. Compute fixation times and drift-escape probabilities for advantages of 2%, 5%, 10% and 30%. What is the smallest improvement your campaign can find?
5. How many parallel lineages, and why?

Worked solution. Mutations per generation are **9.3 × 10^4, 9.3 × 10^6 and 9.3 × 10^8**, so each base is hit 0.02, **2.0** and

200 times. At 10^9 and above **every possible point mutation arises every generation** — ALE is a selection experiment, and a mutagen adds nothing except for multi-mutation events or at small population size.

Transfers: 1:100 gives 6.6 generations per cycle so 500 generations is **75 days**; 1:1000 gives 10 and 50 days, faster but with harsher bottlenecks that discard rare lineages. Fixation needs **1,036 generations at s = 2% and 207 at 10%**, with only **4%** of 2% mutations escaping drift against 20% at 10% — so a few-hundred-generation campaign can realistically only find improvements of **about 5% or more**, and if your objective offers less, ALE is the wrong tool.

Run **6 to 12 parallel lineages**, not one deep one: a passenger appearing in any lineage with probability 0.3 appears in all six with probability 7×10^{-4} and in all twelve with 5×10^{-7}, so parallelism is what separates causal genes from hitchhikers — followed by reconstruction in the clean parent, because parallelism shows association and only reconstruction shows cause.

What it teaches: ALE's design variables are population size, bottleneck and number of lineages, and its resolution limit is a selection coefficient of a few per cent.

`SB4` · chapter Y1 · 1 h

Hypermutators: when to want one

1. At 30× and 100× the wild-type rate, compute the new inactivating mutations per generation in one 1 kb gene across a 10^{16}-cell fermenter.
2. Why is the wild-type number already large, and what does the hypermutator actually change?
3. State the rule for ALE against production.
4. You receive an evolved strain from a collaborator. What do you check first?

Worked solution. 3×10^{11} at 30× and 10^{12} at 100×, against 10^{10} at the wild-type rate. The wild-type figure is already enormous because the population is — part 14 chapter B10's point — so what the hypermutator changes is not whether single mutants exist (they always do) but the supply of **double mutants and of rarer events** such as IS transposition and deletions, which is exactly the class that breaks pathways.

The rule: **a hypermutator is an asset in an ALE campaign and a liability in production**, so repair the mismatch-repair defect before the strain is qualified. On receiving an evolved strain, check first for **mutations in *mutS*, *mutL*, *mutH* or *uvrD*** — many evolved isolates acquire them, and a strain that evolved fast may have done so because it was mutating fast, in which case it will keep mutating in your hands.

What it teaches: mutation rate is itself an evolvable trait, and an evolved strain's speed may be a warning rather than a feature.

`SC1` · chapter Y3 · 1.5 h

Compute your own screening funnel

1. Tabulate the daily throughput of sorting, optical plate screening, HPLC, flask culture and bench bioreactors.
2. Compute the ratio between what a sort delivers in a day and what mass spectrometry verifies.
3. Design a four-tier funnel from 10^7 to 20 strains and compute the instrument-days at each tier.
4. Which tier costs the most time, and which discard worries you most?

Worked solution. Per working day: sorting **5×10^7 cells**, optical screening **3,000 clones**, LC-MS **150 samples**, HPLC **110**, flask plus HPLC **24**, bench bioreactor **4**. A sort delivering 10^5 candidates against 150 verifications a day is a ratio of about **670 to 1**: you cannot verify what you sorted, so the funnel is forced rather than chosen.

The funnel: 10^7 sorted keeping 0.1% → 10,000; optical screen keeping 10% → 1,000; HPLC on 300 keeping 30% → 90; flask and bioreactor on 20. Instrument-days are **0.2, 3.3, 2.7 and 0.8** — so the glamorous tier is the cheapest and **the optical screen is the most expensive in time**. The discard that should worry you is between tiers two and three: 1,000 clones pass the optical screen and 300 go forward, so the proxy assay's ranking is trusted to discard 700, and if the proxy correlates imperfectly with product — which L1 and EV5 both quantify — your best strain is probably among them.

What it teaches: campaign design is arithmetic on assay capacity, and the tier that discards on a proxy is where the best strain is lost.

`SC2` · chapter Y3 · 1.5 h

Where the mass-spec week actually goes

1. Sum the hours per 96-well plate for quench and extraction, dry down, derivatisation, resuspension and a 12-minute instrument method.
2. How many plates can one operator with one instrument manage per working week?
3. What does shortening the method to 3 minutes do, and what does it cost?
4. What does pooling four wells per injection buy and destroy?

Worked solution. 2.0 + 1.5 + 2.0 + 0.5 + 19.2 = **25.2 hours per plate**, so one operator with one instrument manages about **1.6 plates — roughly 150 samples — per week**. Note the structure: the instrument run is 76% of it, so automating the prep helps less than it appears.

Shortening the method from 12 to 3 minutes cuts the run to 4.8 hours and takes the plate to **10.8 hours**, roughly **2.3× the throughput**. It costs chromatographic resolution, so co-eluting peaks — exactly R1's problem — become more likely, and the method must be re-validated for the specific matrix. Pooling four wells per injection multiplies throughput **fourfold** and destroys the identity of three in four wells: it works as a *coarse* tier where you only need to know which pool contains a hit and will re-run its members, and it fails when you need a value per clone. Both are legitimate; both must be stated in a methods section.

What it teaches: the instrument run dominates sample-prep time, so throughput is bought with method length or pooling, each with a specific cost.

`SC3` · chapter Y4 · 1.5 h

Plan a campaign that will be contaminated

1. At per-batch contamination probabilities of 2%, 5% and 10%, compute expected losses over 50 batches and the probability of none.
2. What does that do to a schedule built on 50 successful batches?
3. Compute the cost of a loss relative to a successful batch.
4. Where do plant contaminations usually originate, and what does that imply for where to spend?

Worked solution. Expected losses are **1.0, 2.5 and 5.0**, and the probability of a clean campaign is **0.364, 0.077 and 0.005**. At 5% the chance of 50 clean batches is **8%**, so a schedule built on 50 successes will be missed with 92% probability — the fix is to plan 53 batches, in the spirit of V1's 80th-percentile promise rather than a most-likely one.

A lost batch costs the full medium, the vessel time and the seed train, and it costs them *without* the product that would have paid for them — so a 5% contamination rate is roughly a 5% tax on cost of goods plus the schedule risk. On origin: medium sterilisation is usually the best-controlled step, because it is computed (part 14 chapter B10), while **seals, sample ports, drains and air filters** are the usual culprits. That is where to spend: on seal replacement schedules, sample-port design and air-filter integrity testing, rather than on longer sterilisation cycles that damage the medium for no gain.

What it teaches: contamination is a per-batch probability to be planned for, and the step that is computed is rarely the step that fails.

`SC4` · chapter Y4 · 1.5 h

Tank three is crashing. What is it, and what do you do?

1. Compute the time for a lytic phage to collapse a culture from a free-phage fraction of 10^{-9}, with a burst of 100 and a latent period of 30 minutes.
2. List the three instrument signatures of a phage infection and name the other problem that produces one of them.
3. Why does 0.2 µm filtration not protect you?
4. Rank six defences and name the one you would do before any scale-up.

Worked solution. The collapse time is **about 2.2 hours** from 10^{-9} and 1.8 from 10^{-7} — against a 48-hour fermentation, **a phage entering at inoculation ends the run before the first shift change**.

The signatures are **falling optical density, sharply rising dissolved oxygen as respiration stops, and a clearing broth**. Rising dissolved oxygen during active feeding is also the signature of substrate inhibition (problem X6), which is why the first diagnostic is a **microscope or a plaque assay**, not a feed adjustment — treating a phage infection as an inhibition problem loses the tank and contaminates the plant.

Filtration fails because phage are **30 to 200 nm** and a 0.2 µm membrane passes them; any phage-risk stream needs 0.02 µm filtration or heat. Ranked defences: **screen the production strain for inducible prophage before scale-up**

(first, because it is self-inflicted and cheap to check by sequencing); segregate air, drains and personnel between seed train and production; maintain seals and sample ports; qualify a second strain so rotation is possible; add CRISPR spacers against the local phage, accepting that escape mutants arise; and delete the receptor as a last resort, since it usually costs fitness and selects for phage using another receptor.

What it teaches: a phage infection is faster than any corrective action, so the whole defence is preventive — and the first prevention is sequencing your own strain.

Closing notes to part 22

On the feedback itself

Both criticisms were correct, and unequally so. **The analytics bottleneck was the sharper one**: "samples per day" and "screening funnel" each appeared zero times across twenty-one parts, while FACS throughput and chromatography throughput both appeared and were never divided by one another. That is a clean omission and chapter Y3 is the fix.

The stability criticism was about integration rather than coverage. Plasmid retention, cheater fixation and mutation supply were all present and all separate; what was missing was the product of them along a real seed train, and the two specific mechanisms the audit found thin — IS transposition at six mentions and hypermutators at four. Chapter Y1 integrates them, and the result is a cleaner statement than any of the three gave alone: **batch is safe, repeated batch collapses in three passages, continuous needs growth coupling or integration**.

The addition neither of us had made is contamination and phage. It is unglamorous, it ends real campaigns, and the module had computed sterilisation while never computing what happens when sterility fails.

The numbers worth keeping

- Plasmid-positive fraction falls to **0.951** by the end of the seed train and **0.907** at harvest; cell-weighted, **0.875**.
- Repeated batch: **0.876 → 0.233 → 0.001** over four re-inoculations.
- An IS mutant at an 8% advantage needs **180 generations** to reach half the population.
- At $N_e = 10^9$, **every base in the genome is mutated twice per generation**.
- A 2% beneficial mutation needs **1,036 generations** to fix and has a **4%** chance of escaping drift.
- A passenger appears in all of twelve parallel lineages with probability **5×10^{-7}**.
- Sorting delivers 10^5 candidates a day against **150** mass-spec verifications — **670 to 1**.
- One 96-well plate of GC-MS is **25.2 hours**; about 1.6 plates per operator-week.
- At 5% contamination per batch, the chance of 50 clean batches is **8%**.
- A phage collapses a tank in **2.2 hours** from a 10^{-9} starting fraction.

On the assumptions

All arithmetic computed here. The inputs to challenge: the plasmid loss rate of 5×10^{-4} and the 8% burden (both illustrative, and PP3 says to measure them with a plate count); the IS transposition rate, which varies by element, by host and with stress; the assay throughputs in chapter Y3, which depend heavily on your method and automation and should be replaced with your own laboratory's numbers; and the phage burst size and latent period, which are organism- and phage-specific. The *shapes* of these relations hold; the coefficients are local.

Where the set stands

Source	Problems
Curriculum appendix A1–W3	54
Parts 15–21	85
Part 22 SB1–SB4, SC1–SC4	8
Total in the tracker	**149**